them for all p in some cofinal $Q \subseteq P$. For example, we can construct spanning trees inductively in all the G_n by expanding a dummy vertex in the tree $T_n \subseteq G_n$ to a star in $T_{n+1} \subseteq G_{n+1}$. Then our given bonding maps $X_{p_n} \to X_{p_m}$ will map the subspace X'_{p_n} induced by T_n to that induced by T_m, and these X'_{p_n} will have a topological spanning tree in $X = |G|$ as their inverse limit. This construction is possible only because the partition classes of the p_n are connected in G; we could not perform it on all of $P(G)$.

Arcs and circles in $|G|$, or in a standard subspace, can be obtained easily by applying the following *lifting lemma* with $Y = [0,1]$ or $Y = S^1$. Let $(X_p \mid p \in P)$ be any inverse system of compact spaces, with bonding maps f_{qp} say, and let X be its inverse limit. Let Y be a topological space with continuous *compatible* maps $g_p : Y \to X_p$: maps that commute with the f_{qp} in that $g_p = f_{qp} \circ g_q$ whenever $p < q$. Let us call the family $(g_p \mid p \in P)$ *eventually injective* if for all distinct $y, y' \in Y$ there exists some $p \in P$ with $g_p(y) \neq g_p(y')$.

compatible maps

lifting lemma

Lemma 8.8.4. *There is a unique continuous map $g : Y \to X$ that commutes with the projections $f_p : X \to X_p$ in that $g_p = f_p \circ g$ for all $p \in P$. If the g_p are eventually injective, then g is injective.* □

For example, suppose we wish to find an arc in X between some points x and y. We can find a topological x–y path $g : [0,1] \to X$ by finding topological $f_p(x)$–$f_p(y)$ paths $g_p : [0,1] \to X_p$ that commute with the f_{qp}. If we can make these g_p eventually injective, then g will be injective, and its image will be the desired arc.

Similarly, if we can find compatible circles $g_p : S^1 \to X_p$ that are eventually injective, whose images contain all the vertices of G/p, and which commute with the f_{qp}, then g will define a *Hamilton circle* of G, a circle in $|G|$ that traverses every vertex.

Hamilton circle

Exercises

1.$^-$ Show that a connected graph is countable if all its vertices have countable degrees.

2.$^-$ Given countably many sequences $\sigma^i = s^i_1, s^i_2, \dots$ ($i \in \mathbb{N}$) of natural numbers, find one sequence $\sigma = s_1, s_2, \dots$ that beats every σ^i eventually, i.e. such that for every i there exists an $n(i)$ such that $s_n > s^i_n$ for all $n \geqslant n(i)$.

3.$^-$ Can a countable set have uncountably many subsets whose intersections have finitely bounded size?

4.$^-$ Let T be an infinite rooted tree. Show that every ray in T has an increasing tail, that is, a tail whose sequence of vertices increases in the tree-order associated with T and its root.

5.$^-$ Let G be an infinite graph and $A, B \subseteq V(G)$. Show that if no finite set of vertices separates A from B in G, then G contains an infinite set of disjoint A–B paths.

6.$^-$ In Proposition 8.1.1, the existence of a spanning tree was proved using Zorn's lemma 'from below', to find a maximal acyclic subgraph. For finite graphs, one can also use induction 'from above', to find a minimal spanning connected subgraph. What happens if we apply Zorn's lemma 'from above' to find such a subgraph?

For the next two exercises it may help to consider the cycle space of the given graph, defined as for finite graphs in Chapter 1.9.

7.$^-$ Show that if a graph has a spanning tree with infinitely many chords then all its spanning trees have infinitely many chords.

8. Show that if a graph contains infinitely many distinct cycles then it contains infinitely many edge-disjoint cycles.

9. Let G be a countable infinitely connected graph. Show that G has, for every $k \in \mathbb{N}$, an infinitely connected spanning subgraph of girth at least k.

10. Construct, for any given $k \in \mathbb{N}$, a planar k-connected graph. Can you construct one whose girth is also at least k? Can you construct an infinitely connected planar graph?

11. Theorem 8.1.3 implies that there exists an $\mathbb{N} \to \mathbb{N}$ function f_χ such that, for every $k \in \mathbb{N}$, every infinite graph of chromatic number at least $f_\chi(k)$ has a finite subgraph of chromatic number at least k. (E.g., let f_χ be the identity on $\mathbb{N}$.) Find similar functions f_δ and f_κ for the minimum degree and connectivity, or show that no such functions exist.

12. Let k be an integer and $\kappa \geqslant \aleph_1$ a regular cardinal. Show that every k-connected graph of order κ contains $TK_{k,\kappa}$ as a topological minor.

13. Let $k \in \mathbb{N}$, and let a, b be two vertices in a graph G.
   (i) Show that there are only finitely many minimal a–b separators of order k, and only finitely many minimal a–b cuts of order k. (A cut is an a–b *cut* if it separates a from b.)
   (ii) Deduce that every edge of G lies in only finitely many bonds of k edges.

14.$^+$ Extend Exercise 42 of Chapter 1 to infinite graphs.

15. Use the infinity lemma to show that a rayless connected graph of minimum degree d has a finite subgraph of minimum degree d.

16. A theorem of Halin says that every graph of chromatic number $\alpha \geqslant \aleph_0$ contains a TK^β for every cardinal $\beta < \alpha$. Prove this for $\alpha = \aleph_0$.

17.$^-$ Prove Theorem 8.1.3 for arbitrary graphs using the generalized infinity lemma from Appendix A.

18. Give a proof of Theorem 8.1.3 for countable graphs that is based on the fact that, in this case, the topological space X defined in the third proof of the theorem is sequentially compact. (Thus, every infinite sequence of points in X has a convergent subsequence: there is an $x \in X$ such that every neighbourhood of x contains a tail of the subsequence.)

19.$^-$ Extend Nash-Williams's tree covering theorem (2.4.3) to infinite graphs.

20.$^+$ Extend the packing-covering theorem (2.4.4) to infinite graphs.

21.$^+$ For every $k \in \mathbb{N}$, construct a k-connected locally finite graph such that the deletion of the edge set of any cycle disconnects that graph. Deduce that the tree packing theorem (2.4.1) of Nash-Williams and Tutte fails for infinite graphs.

(Hint. Start with a k-connected finite graph G_0. If G_0 has a cycle C such that deleting $E(C)$ does not disconnect G_0, graft some more copies of G_0 on to $E(C)$ to give C that property. Continue inductively.)

22. Derive the generalized infinity lemma and the compactness principle in Appendix A from each other.

23. In the text, the unfriendly partition conjecture is proved for locally finite graphs, using the infinity lemma.

    (i) Give an alternative proof using the compactness principle from Appendix A.

    (ii) The proof in the text, by the infinity lemma, required a modification of the statement. Is this still necessary? Which step in the proof using the compactness principle reflects the requirement in the infinity lemma that every admissible partial solution must induce an admissible solution on a smaller substructure? Where is the local finiteness used?

24. (i) Prove the unfriendly partition conjecture for countable graphs with all degrees infinite.

    (ii) Can you adapt the proof to cover also those countable graphs that have finitely many vertices of finite degree?

25. Rephrase Gallai's partition theorem of Exercise 55, Ch. 1, in terms of degrees, and extend the equivalent version to locally finite graphs.

26. Prove Theorem 8.4.10 for locally finite graphs. Does your proof extend to arbitrary countable graphs?

27. Extend the marriage theorem to locally finite graphs, but show that it fails for countable graphs with infinite degrees.

28. Show that every locally finite factor-critical graph is finite.

29.$^+$ Show that a locally finite graph G has a 1-factor if and only if, for every finite set $S \subseteq V(G)$, the graph $G - S$ has at most $|S|$ odd (finite) components. Find a counterexample that is not locally finite.

30.$^+$ Extend Kuratowski's theorem to countable graphs.

31.$^-$ A vertex $v \in G$ is said to *dominate* an end ω of G if any of the following three assertions holds; show that they are equivalent.

(i) For some ray $R \in \omega$ there is an infinite v–$(R-v)$ fan in G.

(ii) For every ray $R \in \omega$ there is an infinite v–$(R-v)$ fan in G.

(iii) No finite subset of $V(G-v)$ separates v from a ray in ω.

32. Show that a graph G contains a $TK^{\aleph_0}$ if and only if some end of G is dominated by infinitely many vertices.

33. Let G be a *finitely separable* graph, one in which any two vertices can be separated by finitely many edges.

(i) Show that any two ends in G that cannot be separated by finitely many edges are dominated by a common vertex.

(ii) Is the assumption of finite separability necessary for (i) to hold?

34. Construct a countable graph with uncountably many thick ends. Can you find a locally finite such graph?

35. Show that a locally finite connected vertex-transitive graph has exactly 0, 1, 2 or infinitely many ends.

36.$^+$ Show that the automorphisms of a graph $G = (V, E)$ act naturally on its ends, i.e., that every automorphism $\sigma: V \to V$ can be extended to a map $\sigma: \Omega(G) \to \Omega(G)$ such that $\sigma(R) \in \sigma(\omega)$ whenever R is a ray in an end ω. Prove that, if G is connected, every automorphism σ of G fixes a finite set of vertices or an end. If σ fixes no finite set of vertices, can it fix more than one end? More than two?

37.$^-$ Show that a locally finite spanning tree of a graph G contains a ray from every end of G.

38. A ray in a graph *follows* another ray if the two have infinitely many vertices in common. Show that if T is a normal spanning tree of G then every ray of G follows a unique normal ray of T.

39. Use normal spanning trees to show that a countable connected graph has either countably many or continuum many ends.

40. Show that the following assertions are equivalent for connected countable graphs G.

(i) G has a locally finite spanning tree.

(ii) For no finite separator $X \subseteq V(G)$ in G does $G - X$ have infinitely many components.

41. Show that every (countable) planar 3-connected graph has a locally finite spanning tree.

42. Prove the following infinite version of the Erdős-Pósa theorem: an infinite graph G either contains infinitely many disjoint cycles or it has a finite set Z of vertices such that $G - Z$ is a forest.

43. Let G be a connected graph. Call a set $U \subseteq V(G)$ *dispersed* if every ray in G can be separated from U by a finite set of vertices. (In the topology of Section 8.6, these are precisely the closed subsets of $V(G)$.)
    (i) Prove that G has a normal spanning tree if and only if $V(G)$ is a countable union of dispersed sets. Show that in this case we can choose any vertex of G as the normal spanning tree's root.
    (ii) Deduce that if G has a normal spanning tree then so does every connected minor of G.

44.$^+$ Show that a connected graph G has a normal spanning tree if it does not contain a subdivision of a *fat* $K^{\aleph_0}$, one in which every edge has been replaced by uncountably many parallel edges.

    (Hint. Apply induction on $|G|$. Given G, construct an increasing sequence $(\, G_\beta \mid \beta < \alpha \,)$ of smaller graphs exhausting G so that the endvertices of any G_β-path in G are joined in G_β by infinitely many independent paths and in G by uncountably many independent paths. Combine suitable normal spanning trees of the components of $G_{\beta+1} - G_\beta$ with the inductively given normal spanning tree of G_β to form a normal spanning tree of $G_{\beta+1}$.)

45.$^+$ Prove Theorem 8.2.5 (ii).

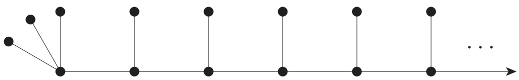
46. (i) Prove that if a given end of a graph contains k disjoint rays for every $k \in \mathbb{N}$ then it contains infinitely many disjoint rays.

    (ii)$^+$ Prove that if a given end of a graph contains k edge-disjoint rays for every $k \in \mathbb{N}$ then it contains infinitely many edge-disjoint rays.

47. Prove that if a graph contains k disjoint double rays for every $k \in \mathbb{N}$ then it contains infinitely many disjoint double rays.

48. Show that, in the ubiquity conjecture, the host graphs G considered can be assumed to be locally finite too.

49. Show that the modified comb below is not ubiquitous with respect to the subgraph relation. Does it become ubiquitous if we delete its 3-star on the left?



50. Show that every locally finite tree T is minor-ubiquitous in countable graphs, by proving and combining the following statements:
    (i) The $\mathbb{N} \times \mathbb{N}$ grid H satisfies $\aleph_0 H \preccurlyeq H$.
    (ii) $T \preccurlyeq H$.
    (iii)$^+$ Assuming that $nT \preccurlyeq G$ for all $n \in \mathbb{N}$, show that $\aleph_0 T \preccurlyeq G$ unless G has a thick end.

51. Imitate the proof of Theorem 8.2.6 to find a function $f\colon \mathbb{N} \to \mathbb{N}$ such that whenever an end ω of a graph G contains $f(k)$ disjoint rays, G contains a subdivision of the $k \times \mathbb{N}$ hexagonal grid whose rays all belong to ω.

52. Show that there is no universal locally finite connected graph for the subgraph relation.

53. Construct a universal locally finite connected graph for the minor relation. Is there one for the topological minor relation?

54.$^-$ Show that each of the following operations performed on the Rado graph R leaves a graph isomorphic to R:

(i) taking the complement, i.e. changing all edges into non-edges and vice versa;

(ii) deleting finitely many vertices;

(iii) changing finitely many edges into non-edges or vice versa;

(iv) changing all the edges between a finite vertex set $X \subseteq V(R)$ and its complement $V(R) \setminus X$ into non-edges, and vice versa.

55. (i)$^-$ If we delete infinitely many vertices of the Rado graph R leaving an infinite rest, does the rest necessarily induce a copy of R?

(ii) Does R admit a vertex partition into two copies of itself?

(iii) Construct a locally finite tree that has a vertex partition into two copies of itself.

56.$^-$ Prove that the Rado graph is homogeneous.

57. Show that a homogeneous countable graph is determined uniquely, up to isomorphism, by the class of (the isomorphism types of) its finite subgraphs.

58. Recall that subgraphs $H_1, H_2, \ldots$ of a graph G are said to *partition* G if their edge sets form a partition of $E(G)$. Show that the Rado graph can be partitioned into any given countable set of countable locally finite graphs, as long as each of them contains at least one edge.

59. A linear order is called *dense* if between any two elements there lies a third.

(i) Find, or construct, a countable dense linear order that has neither a maximal nor a minimal element.

(ii) Show that this order is unique, i.e. that every two such orders are order-isomorphic. (Definition?)

(iii) Show that this ordering is universal among the countable linear orders. Is it homogeneous? (Supply appropriate definitions.)

60. Given a bijection f between $\mathbb{N}$ and $[\mathbb{N}]^{<\omega}$, let G_f be the graph on $\mathbb{N}$ in which $u, v \in \mathbb{N}$ are adjacent if $u \in f(v)$ or vice versa. Prove that all such graphs G_f are isomorphic.

61. (for set theorists) Show that, given any countable model of set theory, the graph whose vertices are the sets and in which two sets are adjacent if and only if one contains the other as an element, is the Rado graph.

62. Find two proofs that every infinitely edge-connected graph has infinitely many edge-disjoint spanning trees:

    (i)$^-$ Apply Theorem 8.4.1.

    (ii)$^+$ Find a direct argument.

63.$^-$ Given sets A, B of vertices in a graph G, show that either G contains infinitely many edge-disjoint A–B paths or there is a finite set of edges separating A from B in G.

64. Let G be a locally finite graph. Let us say that a finite set S of vertices *separates* two ends ω and ω' if $C(S,\omega) \neq C(S,\omega')$. Use Proposition 8.4.3 to show that if ω can be separated from ω' by $k \in \mathbb{N}$ but no fewer vertices, then G contains k disjoint double rays each with one tail in ω and one in ω'. Is the same true for all graphs that are not locally finite?

65.$^+$ Prove the following more structural version of Exercise 46 (i). Let ω be an end of a graph G. Show that either G contains a $TK^{\aleph_0}$ with all its rays in ω, or there are disjoint finite sets $S_0, S_1, \dots$ such that, if C_i is the component of $G - (S_0 \cup S_i)$ that contains a tail of every ray in ω, we have for all $i < j$ that $C_i \supsetneq C_j$ and $G[S_i \cup C_i]$ contains $|S_i|$ disjoint S_i–S_{i+1} paths for all $i \geqslant 1$.

66.$^+$ Is there a planar $\aleph_0$-regular graph all whose ends have infinite vertex-degree?

67.$^-$ Let A, B be two vertex sets in a locally finite connected graph G. Can there be an infinite sequence $\mathcal{P}_1, \mathcal{P}_2, \dots$ of disjoint A–B paths such that each $\mathcal{P}_{n+1}$ arises from $\mathcal{P}_n$ by applying an alternating walk, and such that some edge $e \in G$ lies in $E[\mathcal{P}_n]$ for infinitely many n but not in $E[\mathcal{P}_n]$ for infinitely many other n?

68. Construct an example of a small limit of large waves. Can you find a locally finite one?

69.$^+$ Prove Theorem 8.4.4 for trees.

70.$^+$ Prove Pym's theorem (8.4.9).

71. (i)$^-$ Prove the naive extension of Dilworth's theorem to arbitrary infinite posets P: if P has no antichain of order $k \in \mathbb{N}$, then P can be partitioned into fewer than k chains. (A proof for countable P will do.)

    (ii)$^-$ Find a poset that has no infinite antichain and no partition into finitely many chains.

    (iii) For posets without infinite chains, deduce from Theorem 8.4.10 the following Erdős-Menger-type extension of Dilworth's theorem: every such poset has a partition $\mathcal{C}$ into chains such that some antichain meets all the chains in $\mathcal{C}$.

72. Let G be a countable graph in which for every partial matching there is an augmenting path.

    (i) Find an example of G and a sequence $M_0, M_1, \dots$ of partial matchings, each obtained from the previous as its symmetric difference with the edge set of an augmenting path, so that for every edge e of G we have $e \in M_{n+1} \setminus M_n$ for infinitely many n.

(ii) Show that for every partial matching M there exists a sequence as in (i) such that $\bigcup_m \bigcap_{n>m} M_n$ is the edge set of a 1-factor.

73. Find an uncountable graph in which every partial matching admits an augmenting path (finite or infinite) but which has no 1-factor.

74.$^-$ Let G be a countable graph whose finite subgraphs are all perfect. Show that G is weakly perfect but not necessarily perfect.

75.$^+$ Let G be the incomparability graph of the binary tree. (Thus, $V(G) = V(T_2)$, and two vertices are adjacent if and only if they are incomparable in the tree-order of T_2.) Show that G is perfect but not strongly perfect.

76.$^+$ (i) Show that the vertices of any infinite connected locally finite graph can be enumerated in such a way that every vertex is adjacent to some later vertex.

(ii) Characterize the class of all these graphs, countable but not necessarily locally finite, by their separation properties.

77. Show that a tree has a rank as defined in the second paragraph after the proof of Proposition 8.5.1 if and only if it is recursively prunable, and that it has rank α if and only if α is the maximum of its pruning labels.

78. Let G be a rayless graph, of rank α say, and let U be a finite set of vertices witnessing this, of minimal order. Show that U is unique.

79. (i) Construct a countable tree that has rank ω in the ranking of rayless graphs. Can you find one such tree that contains all the others?

(ii)$^+$ Is there a tree of rank ω that is a subtree of every such tree?

80. A graph $G = (V, E)$ is called *bounded* if for every vertex labelling $\ell : V \to \mathbb{N}$ there exists a function $f : \mathbb{N} \to \mathbb{N}$ that exceeds the labelling along any ray in G eventually. (Formally: for every ray $v_1 v_2 \dots$ in G there exists an n_0 such that $f(n) > \ell(v_n)$ for every $n > n_0$.) Prove the following assertions:

(i) The ray is bounded.

(ii) Every locally finite connected graph is bounded.

(iii) A countable tree is bounded if and only if it contains no subdivision of the $\aleph_0$-regular tree $T_{\aleph_0}$.

81.$^+$ Let T be a tree with root r, and let $\leqslant$ denote the tree-order on $V(T)$ associated with T and r. Show that T contains no subdivision of the $\aleph_1$-regular tree $T_{\aleph_1}$ if and only if T has an ordinal labelling $t \mapsto o(t)$ such that $o(t) \geqslant o(t')$ whenever $t < t'$ and no more than countably many vertices of T have the same label.

82. Let G be a countable connected graph with vertices $v_0, v_1, \dots$. For every $n \in \mathbb{N}$ write $S_n := \{v_0, \dots, v_{n-1}\}$. Prove the following statements:

(i) For every end ω of G there is a unique sequence $C_0 \supseteq C_1 \supseteq \dots$ of components C_n of $G - S_n$ such that $C_n = C(S_n, \omega)$ for all n.

(ii) For every infinite sequence $C_0 \supseteq C_1 \supseteq \ldots$ of components C_n of $G - S_n$ there exists a unique end ω such that $C_n = C(S_n, \omega)$ for all n.

83. Let G be a graph, $U \subseteq V(G)$, and $R \in \omega \in \Omega(G)$. Show that G contains a comb with spine R and teeth in U if and only if $\omega \in \overline{U}$.

84. Given graphs $H \subseteq G$, let $\eta \colon \Omega(H) \to \Omega(G)$ assign to every end of H the unique end of G containing it as a subset (of rays). For the following questions, assume that H is connected and $V(H) = V(G)$.

    (i) Show that η need not be injective. Must it be surjective?

    (ii) Investigate how η relates the subspace $\Omega(H)$ of $|H|$ to its image in $|G|$. Is η always continuous? Is it open? Do the answers to these questions change if η is known to be injective?

    (iii) A spanning tree is called *end-faithful* if η is bijective, and *topologically end-faithful* if η is a homeomorphism. Show that every connected countable graph has a topologically end-faithful spanning tree.

The *end space* of a graph G is the subspace $\Omega(G)$ of $|G|$.

85. Consider the end space Ω of the binary tree T_2 shown in Figure 8.1.4, in which its vertices are the finite 0–1 sequences.

    (i) Show that Ω is homeomorphic to $2^{\mathbb{N}}$, where $2 = \{0, 1\}$ carries the discrete topology and $2^{\mathbb{N}}$ the product topology.

    (ii) Identify in Ω every two ends whose infinite binary sequences encode the same rational. Show that the resulting quotient space of Ω is homeomorphic to the real interval $[0, 1]$.

86. Above every horizontal edge of the plane graph shown in Figure 8.6.1 add infinitely many horizontal edges in the plane, so as to turn every pair of rays whose associated 0–1 sequences define the same rational number into a ladder. Prove or disprove that the end space of the resulting graph is homeomorphic to $[0, 1]$.

87. A compact metric space is a *Cantor set* if the singletons are its only connected subsets and every point is an accumulation point.

    (i) Characterize the trees whose end space is a Cantor set.

    (ii) Show that the end space of a connected locally finite graph is a subset of a Cantor set.

88. (i) Show that if H is a contraction minor of G with finite branch sets, then the end spaces of H and G are homeomorphic.

    (ii) Let T_n denote the n-ary tree, the rooted tree in which every vertex has exactly n successors. Show that all these trees have homeomorphic end spaces.

89. Give an independent proof of Proposition 8.6.1 using sequential compactness and the infinity lemma.